ECOM90009 Quantitative Methods for Business

Workshop 7 (Week 8)

Zheng Fan

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Assignment 2

Assignment 2 is now available on Canvas. This assignment contributes up to 10% of your final grade for this subject.

Submission Deadline: Thursday, May 7 at 11:59 pm.

Important: You must register again your group by Thursday, April 30 at 11:59 pm.

- The maximum group size is 4.
- Group members do not need to be the same as those in Assignment 1.

Introduction

- Hypothesis test involving comparison of two populations
  - Matched pairs:
    `Matched pair samples use the same members in both samples.`
  - Independent samples:
    `Two samples that are completely unrelated to each other are said to be independent.`
- Test of equality of two population variances

Question 1 (Matched pair or independent samples)

State whether you would describe each of the following as matched pair or independent samples. Explain your answers.

(a) You randomly select a group of companies and observe the profits they make before and after a carbon emission trading scheme is introduced.

Question 1 (Matched pair or independent samples)

State whether you would describe each of the following as matched pair or independent samples. Explain your answers.

(a) You randomly select a group of companies and observe the profits they make before and after a carbon emission trading scheme is introduced.

- Same companies are observed before and after a carbon emissions trading scheme is introduced
- Matched pairs

Question 1 (Matched pair or independent samples)

(b) You compare the average time customers are kept waiting before being seated at restaurants in Melbourne's central business district with waiting times at restaurants in Carlton.

Question 1 (Matched pair or independent samples)

(b) You compare the average time customers are kept waiting before being seated at restaurants in Melbourne's central business district with waiting times at restaurants in Carlton.

- The restaurants included in each sample are different.
- They cannot be the same restaurants as they are located in different suburbs.
- Independent samples

Question 1 (Matched pair or independent samples)

(c) You record the running time of 50 athletes running a distance of 100m, first wearing one brand of running shoes, and again wearing another brand.

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- The athletes observed in each sample are the same
- Matched pairs

Question 1 (Matched pair or independent samples)

(d) You want to know whether people are more likely to go see a movie if they have seen the movie trailer beforehand. You randomly select a group of people who have seen a particular movie trailer, and observe the proportion that intend to see that movie at the cinema. You then randomly select a group of people from among those who have not seen the trailer for that particular movie, and observe the proportion that intend to see that movie at the cinema.

(e) You measure the life span (in years) of ant colonies in the desert compared to ant colonies in the jungle.

(f) You measure the number of calves a pod of whales has when they are in cold climates, as opposed to when the pod has migrated to warmer climates.

Question 1 (Matched pair or independent samples)

(d) You want to know whether people are more likely to go see a movie if they have seen the movie trailer beforehand. You randomly select a group of people who have seen a particular movie trailer, and observe the proportion that intend to see that movie at the cinema. You then randomly select a group of people from among those who have not seen the trailer for that particular movie, and observe the proportion that intend to see that movie at the cinema.

Question 1 (Matched pair or independent samples)

(d) You want to know whether people are more likely to go see a movie if they have seen the movie trailer beforehand. You randomly select a group of people who have seen a particular movie trailer, and observe the proportion that intend to see that movie at the cinema. You then randomly select a group of people from among those who have not seen the trailer for that particular movie, and observe the proportion that intend to see that movie at the cinema.

- Two randomly selected groups: one that has seen the movie trailer, and the other has not seen the movie trailer.
- People cannot both see a movie trailer, and not see a movie trailer.
- Independent samples

Question 1 (Matched pair or independent samples)

(e) You measure the life span (in years) of ant colonies in the desert compared to ant colonies in the jungle.

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(e) You measure the life span (in years) of ant colonies in the desert compared to ant colonies in the jungle.

- Ant colonies in the desert and jungle are different.
- Independent samples

Question 1 (Matched pair or independent samples)

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(f) You measure the number of calves a pod of whales has when they are in cold climates, as opposed to when the pod has migrated to warmer climates.

- The same pod of whales are observed in each sample.
- Matched pairs

A Summary - page 49 Seminar 7

1. **Matched Pairs Samples**

   - We estimate and test the mean difference μ_D using $\bar{X}_D$ and the t-statistic (variance unknown).
     Require: The differences X_D must be normally distributed, or approximately so.

A Summary - page 49 Seminar 7

1. Matched Pairs Samples

- We estimate and test the mean difference μ_D using $\bar{X}_D$ and the t-statistic (variance unknown).
  Require: The differences X_D must be normally distributed, or approximately so.

2. Independent Samples

We estimate and test the difference in means $\mu_1 - \mu_2$ using $\bar{X}_1 - \bar{X}_2$.

(a) If σ_1^2 and σ_2^2 are known, use the z-score.
Require: X_1 and X_2 are normally distributed, or $n_1 \geq 30$ and $n_2 \geq 30$.

(b) If σ_1^2 and σ_2^2 are unknown but equal, use the t-statistic with a pooled estimate of the sample variance.

(c) If σ_1^2 and σ_2^2 are unknown and not equal, use the t-statistic with separate estimates of sample variances.
Require: for (b) and (c): X_1 and X_2 are both normally distributed, or approximately so.

Question 2

In the Excel spreadsheet `Workshop07.xlsx`, there is data on the number of cars that travel on a road every day for approximately 3 years. In the sheet labelled `Q2 weekend`, there is the variable `n_cars_weekend` that lists the number of cars recorded for each Saturday and Sunday. In the sheet labelled `Q2 weekday` is the variable `n_cars_weekday` that lists the number of cars recorded for each weekday.

Use this data to test the hypothesis that the number of cars is the same for the weekdays and weekends on average.

Comment and/or test on any assumptions that are needed. Perform all tests using the 5% level of significance.

First, matched pair or independent samples?

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First, matched pair or independent samples?

- Sampling scheme: Independent samples

Question 2

Let X_1 be the number of cars that travel on a road on **weekdays**

Let X_2 be the number of cars that travel on a road on **weekends**

Can we use the independent samples t-test?

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Can we use the independent samples t-test?

- Need to check for normality
  - Use Excel to construct histogram

If normality assumption is satisfied and that we proceed with the independent samples t-test, which t-test should be performed?

Question 2

Let X_1 be the number of cars that travel on a road on **weekdays**

Let X_2 be the number of cars that travel on a road on **weekends**

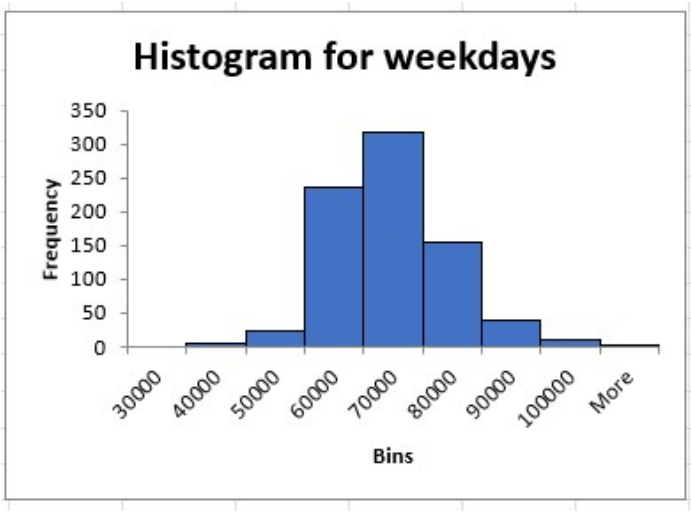
Can we use the independent samples t-test?

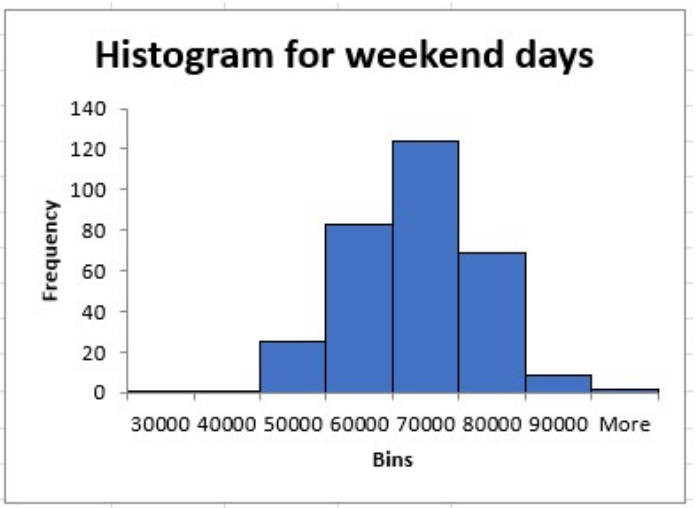
- Need to check for normality
  - Use Excel to construct histogram

If normality assumption is satisfied and that we proceed with the independent samples t-test, which t-test should be performed?

- Since the population variances are unknown, we need to perform a test for whether $\sigma_1^2 = \sigma_2^2$.

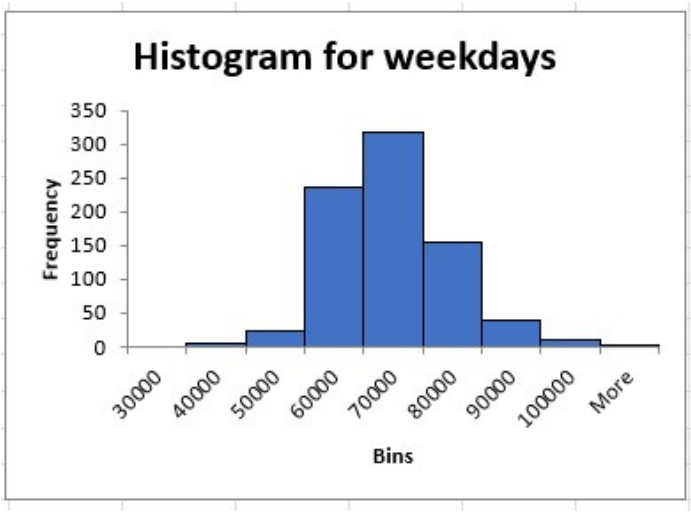
Checking for normality

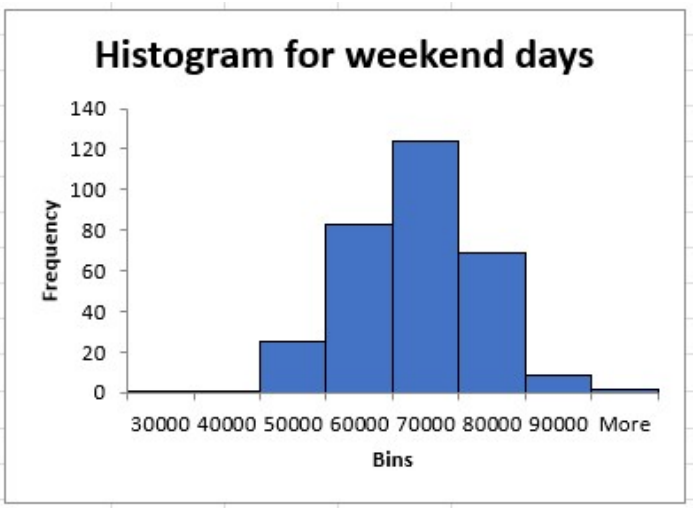




What do the histograms suggest about normality?

Checking for normality





What do the histograms suggest about normality?

- In both cases, the histograms look relatively bell-shaped although the distribution for weekdays might be slightly skewed.
- However, in both cases the samples sizes are large.
- We are going to assume the two populations are approximately normally distributed.

Obtaining sample statistics

N_Cars_weekday	
Mean	64712.94522
Standard Error	349.3692637
Median	63646
Mode	54070
Standard Deviation	9788.576133
Sample Variance	95816222.71
Kurtosis	0.744766411
Skewness	0.564652139
Range	69416
Minimum	35839
Maximum	105255
Sum	50799662
Count	785

N_CARS_weekend	
Mean	63380.21338
Standard Error	555.2901567
Median	63412
Mode	53897
Standard Deviation	9839.766646
Sample Variance	96821007.64
Kurtosis	0.425217628
Skewness	-0.150540605
Range	68365
Minimum	26057
Maximum	94422
Sum	19901387
Count	314

$$\bar{x}_1 = 64712.945, \quad s_1 = 9788.576, \quad s_1^2 = 95816222.71, \quad n_1 = 785$$

$$\bar{x}_1 = 63380.2134, \quad s_2 = 9839.767, \quad s_2^2 = 96821007.64, \quad n_2 = 314$$

Hypothesis test for $\sigma_1^2 = \sigma_2^2$

Step 1: $H_0 : \sigma_1^2/\sigma_2^2 = 1 \qquad H_A : \sigma_1^2/\sigma_2^2 \neq 1$

Step 2: Assuming the two populations are approximately normally distributed, we use:

$$F = \frac{\frac{(n_1-1)s_1^2/\sigma_1^2}{(n_1-1)}}{\frac{(n_2-1)s_2^2/\sigma_2^2}{(n_2-1)}} = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} = \frac{s_1^2}{s_2^2} \quad F \sim F_{n_1-1,n_2-1}$$

where $n_1 - 1 = 784$ and $n_2 - 1 = 313$

Step 3: Use $\alpha = 0.05$

Step 4: Decision rule: Reject H_0 if F-statistic is below $F_{0.975,\ 784,\ 313} = 0.834$ **OR** above $F_{0.025,\ 784,\ 313} = 1.208$

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Step 4: Decision rule: Reject H_0 if F-statistic is below $F_{0.975,\ 784,\ 313} = 0.834$ **OR** above $F_{0.025,\ 784,\ 313} = 1.208$

Excel commands:

- `=F.INV.RT(0.025,784,313)`
- `=F.INV(0.025,784,313)`

Hypothesis test for $\sigma_1^2 = \sigma_2^2$

Step 5: Test statistic calculation

$$F = \frac{s_1^2}{s_2^2} = \frac{95816222.71}{96821007.64} = 0.9896$$

Hypothesis test for $\sigma_1^2 = \sigma_2^2$

Step 5: Test statistic calculation

$$F = \frac{s_1^2}{s_2^2} = \frac{95816222.71}{96821007.64} = 0.9896$$

Step 6: 0.9896 is in between the lower and upper critical values, i.e. does not lie in the rejection regions.

So, we do not reject H_0 at the 5% level of significance.

There is insufficient evidence at the 5% level to support the hypothesis and hence we will assume that the two population variances are equal.

Use the samples t-test for

`Case 1: Variances Unknown but Equal - page 43 Seminar 7`

Test for equality of means

Step 1: Is the number of cars the same for weekdays and weekends on average?

$$H_0 : \mu_1 - \mu_2 = 0 \qquad H_A : \mu_1 - \mu_2 \neq 0$$

Step 2: Independent samples. Assuming the two populations are approximately normally distributed, and equal population variances:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1+n_2-2}$$

where s_p^2 is the pooled estimate of a common population variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Test for equality of means

Step 3: Use $\alpha = 0.05$

Step 4: 2-sided test. Reject H_0 if $t < -1.96$ or $t > 1.96$

Excel commands:

- `=T.INV.2T(0.05,1097)` which gives `1.962129` OR
- `=T.INV(0.025,1097)` which gives the left-tail critical value of `-1.96213`

Test for equality of means

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Step 5: Calculation of test statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(64,712.945 - 63,380.2134)}{\sqrt{96,102,910.86\left(\frac{1}{785} + \frac{1}{314}\right)}} = 2.04$$

where $s_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2} = \frac{784\times 95816222.71+313\times 96821007.64}{785+314-2} = 96,102,910.86$

Step 6: As $t = 2.04 > 1.96$, we reject H_0 at the 5% level of significance.

Question 3

In the Excel spreadsheet `Workshop07.xlsx`, there is data in the sheet labelled `Q3` on car registrations for 477 Postal Code Areas (PCAs) (local areas) in Australia that are located in the major capital cities. In particular, we have data on the number of BMWs and the number of Daihatsu's for each PCA. Use this data to test whether the number of Daihatsu's is the same as the number of BMWs on average. Comment and/or test on any assumptions that are needed. Perform all tests using the 5% level of significance.

First, matched pair or independent samples?

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First, matched pair or independent samples?

- Sampling scheme: Matched pairs

Question 3

Let X_1 denote the number of Daihatsu registrations

Let X_2 denote the number of BMW registrations

Let $X_D = X_1 - X_2$

How do we perform the appropriate test? Matched-pairs t test?

Question 3

Let X_1 denote the number of Daihatsu registrations

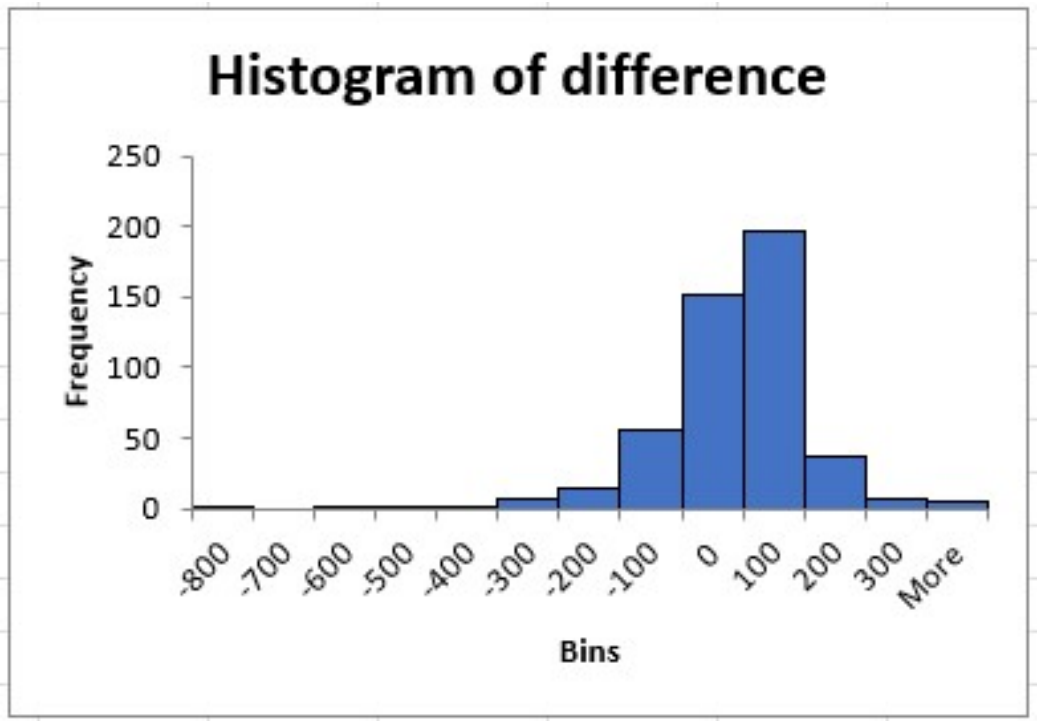
Let X_2 denote the number of BMW registrations

Let $X_D = X_1 - X_2$

How do we perform the appropriate test? Matched-pairs t test?

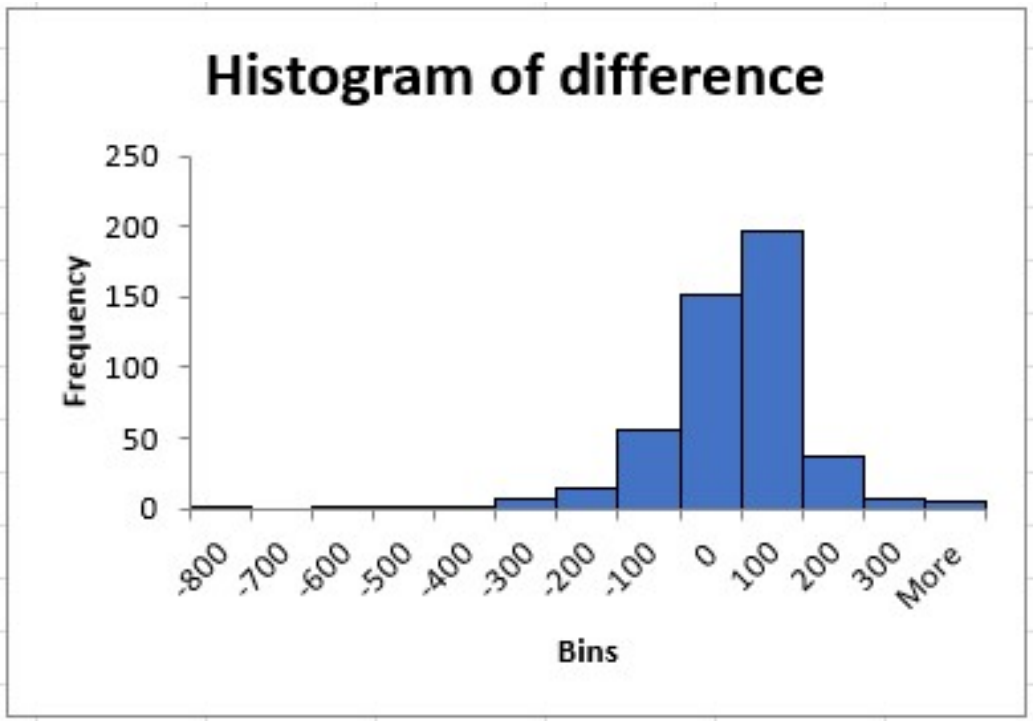
- Need to check for normality of the differences
  - Use Excel to construct histogram

Checking for normality



What does the histogram suggest about normality?

Checking for normality



What does the histogram suggest about normality?

- The data looks quite skewed.

However, we have a large sample consisting of 477 so the assumption that the difference is approximately normally distributed is reasonable.

Obtaining sample statistics

Diff	
Mean	-16.11740042
Standard Error	5.795478593
Median	3
Mode	14
Standard Deviation	126.5751631
Sample Variance	16021.2719
Kurtosis	8.368268237
Skewness	-1.732231749
Range	1218
Minimum	-873
Maximum	345
Sum	-7688
Count	477

$$\bar{x}_D = -16.1174, \quad s_D = 126.5752, \quad n_D = 477$$

Test for equality of means

Step 1: Is the number of car registrations the same on average for Daihatsus and BMWs?

$$H_0 : \mu_D = 0 \qquad H_A : \mu_D \neq 0$$

Step 2: Matched-pairs. Assuming that it is reasonable that the differences are approximately normally distributed, we use:

$$t = \frac{\bar{x}_D - \mu_D}{s_D / \sqrt{n_D}} \sim t_{n_D - 1}$$

Test for equality of means

Step 3: Use $\alpha = 0.05$

Step 4: 2-sided test. Reject H_0 if $t < -1.96$ or $t > 1.96$

Excel commands:

- `=T.INV.2T(0.05,476)` which gives `1.96496` OR
- `=T.INV(0.025,476)` which gives the left-tail critical value of `-1.96496`

Test for equality of means

Step 3: Use $\alpha = 0.05$

Step 4: 2-sided test. Reject H_0 if $t < -1.96$ or $t > 1.96$

Excel commands:

- `=T.INV.2T(0.05,476)` which gives `1.96496` OR
- `=T.INV(0.025,476)` which gives the left-tail critical value of `-1.96496`

Step 5: Calculation of test statistic

$$t = \frac{\bar{x}_D - \mu_D}{s_D/\sqrt{n_D}} = \frac{-16.1174}{126.575/\sqrt{477}} = -2.78$$

Step 6: As $t = -2.78 < -1.96$, we reject H_0 at the 5% level of significance.

There is sufficient evidence at the 5% level to support the alternative hypothesis that there is a difference between the **mean** number of Daihatsu

Thanks for your attention!

Let me know if there is any questions!